

$$\text{Bölm 12} = \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) R(r) + \left\{ \frac{2\mu}{\hbar^2} \left(-\frac{l(l+1)\hbar^2}{2\mu r^2} + V(r) \right) + \frac{2\mu E}{\hbar^2} \right\} R(r) = 0$$

$$\left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) R(r) + \frac{2\mu}{\hbar^2} \left(-\frac{l(l+1)\hbar^2}{2\mu r^2} + \underbrace{V(r)}_{+\frac{ze^2}{r}} + \underbrace{|E|}_{-E} \right) R(r) = 0$$

$$\Rightarrow \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) R(r) + \frac{2\mu}{\hbar^2} \left(-\frac{l(l+1)\hbar^2}{2\mu r^2} + \frac{ze^2}{r} - E \right) R(r) = 0 \quad \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} r = \rho$$

$$\Rightarrow \frac{8\mu E}{\hbar^2} \left(\frac{d}{d\rho^2} + \frac{2}{\rho} \frac{d}{d\rho} \right) R(\rho) + \frac{2\mu}{\hbar^2} \left(-\frac{l(l+1)}{2\mu \rho^2} \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} + \frac{ze^2}{\rho} \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} - E \right) R(\rho) = 0$$

$$\left(\frac{\hbar^2}{8\mu E} \right)^{1/2} \frac{d}{dr} = \frac{d}{d\rho} \quad \frac{d}{dr} = \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} \frac{d}{d\rho}$$

$$\Rightarrow \frac{8\mu E}{\hbar^2} \left\{ \left(\frac{d^2}{d\rho^2} + \frac{2}{\rho} \frac{d}{d\rho} \right) R(\rho) + \left(-\frac{l(l+1)}{\rho^2} \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} + \frac{2\mu}{\hbar^2} \frac{ze^2}{\rho} \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} - \frac{2\mu E}{\hbar^2} \right) R(\rho) \right\} = 0$$

$\forall$ terisi $\frac{8\mu E}{\hbar^2}$ bölmü;

$$\left(\frac{d^2}{d\rho^2} + \frac{2}{\rho} \frac{d}{d\rho} \right) R(\rho) + \left(-\frac{l(l+1)}{\rho^2} + \frac{2\mu}{\hbar^2} \frac{ze^2}{\rho} \left(\frac{\hbar^2}{8\mu E} \right)^{1/2} - \frac{2\mu E}{\hbar^2} \left(\frac{\hbar^2}{8\mu E} \right)^{1/2} \right) R(\rho) = 0$$

$$\left(\frac{d^2}{d\rho^2} + \frac{2}{\rho} \frac{d}{d\rho} \right) R(\rho) + \left(-\frac{l(l+1)}{\rho^2} + \frac{ze^2}{\rho \hbar} \left(\frac{\hbar}{2E} \right)^{1/2} - \frac{1}{4} \right) R(\rho) = 0 \quad \star$$

$$\underbrace{\frac{ze^2}{\rho \hbar} \left(\frac{\hbar}{2E} \right)^{1/2}}_{\frac{N}{\rho}}$$

Asymt Dərrədlis

$$\frac{d}{dr} = \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} \frac{d}{d\rho} \Rightarrow \rho \rightarrow \infty \Rightarrow r \rightarrow 0 \rightarrow \infty$$

$$\frac{d^2}{d\rho^2} - \frac{1}{4} = 0 \Rightarrow D^2 - \frac{1}{4} = 0 \Rightarrow D = \pm \frac{1}{2} \Rightarrow D = e^{\pm \rho/2} \Rightarrow D = e^{-\rho/2}$$

$$\Rightarrow \text{G.G. } D = G(\rho) e^{-\rho/2}$$

$$R(\rho) = G(\rho) e^{-\rho/2}$$

$$R'(\rho) = G'(\rho) e^{-\rho/2} - \frac{1}{2} G(\rho) e^{-\rho/2}$$

$$R''(\rho) = G''(\rho) e^{-\rho/2} - 1 \cdot G'(\rho) e^{-\rho/2} + \frac{1}{4} G(\rho) e^{-\rho/2}$$

$$\Rightarrow R(\rho) = e^{-\rho/2} \{ G \} \quad R' = e^{-\rho/2} \left(G' - \frac{1}{2} G \right) \quad R'' = e^{-\rho/2} \left(G'' - G' + \frac{1}{4} G \right)$$

$$\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \left(R(p) \right) + \left(-\frac{l(l+1)}{p^2} + \frac{\lambda}{p} - \frac{1}{4} \right) R(p) = 0 \quad \star$$

$$R(p) = e^{-p/2} \{ G \} \quad R' = e^{-p/2} \left(G' - \frac{1}{2} G \right) \quad R'' = e^{-p/2} \left(G'' - G' + \frac{1}{4} G \right)$$

$$e^{-p/2} \left\{ \left(G'' - G' + \frac{1}{4} G \right) + \frac{2}{p} \left(G' - \frac{1}{2} G \right) - \frac{l(l+1)}{p^2} G + \frac{\lambda}{p} G - \frac{1}{4} G \right\} = 0$$

$$G'' + G' \left(\frac{2}{p} - 1 \right) + G \left(\frac{1}{4} - \frac{1}{4} - \frac{1}{p} - \frac{l(l+1)}{p^2} + \frac{\lambda}{p} \right) = 0$$

$$\Rightarrow G'' + G' \left(\frac{2}{p} - 1 \right) + G \left(\frac{\lambda - 1}{p} - \frac{l(l+1)}{p^2} \right) = 0 \quad \star \star$$

probeklausur

$$(i) G = H(p) p^l \quad (ii) H = \sum_{n=0}^{\infty} a_n p^n$$

$$G = H p^l \quad G' = H' p^l + l H p^{l-1} \quad G'' = H'' p^l + 2l H' p^{l-1} + l(l-1) H p^{l-2}$$

$$\Rightarrow p^l \left\{ H'' + \frac{2l}{p} H' + \frac{l(l-1)}{p^2} H + \left(\frac{2}{p} - 1 \right) (H' + \frac{l}{p} H) + H \left(\frac{\lambda - 1}{p} - \frac{l(l+1)}{p^2} \right) \right\} = 0$$

$$H'' + H' \left(\frac{2l}{p} + \frac{2}{p} - 1 \right) + H \left(\frac{l(l-1)}{p^2} + \frac{2l}{p^2} - \frac{l}{p} + \frac{\lambda - 1}{p} - \frac{l(l+1)}{p^2} \right) = 0$$

$$H'' + H' \left(\frac{2(l+1)}{p} - 1 \right) + H \left(\frac{\lambda - 1}{p} - \frac{l(l+1)}{p^2} \right) = 0$$

$$H'' + H' \left(\frac{2(l+1)}{p} - 1 \right) + H \left(\frac{\lambda - (l+1)}{p} \right) = 0$$

$$H = \sum_{n=0}^{\infty} a_n p^n$$

$$H' = \sum_{n=0}^{\infty} a_n n p^{n-1}$$

$$\Rightarrow \sum_{n=0}^{\infty} a_n n(n-1) p^{n-2} + \sum_{n=0}^{\infty} a_n n p^{n-1} \left(\frac{2(l+1)}{p} - 1 \right) + \sum_{n=0}^{\infty} a_n p^n \left(\frac{\lambda - (l+1)}{p} \right) = 0$$

$$\Rightarrow \sum_{n+1=1}^{\infty} a_{n+1} (n+1)n p^{n-1} + \sum_{n+1=1}^{\infty} a_{n+1} (n+1) p^{n-1} (2(l+1)) + \sum_{n=0}^{\infty} a_n p^{n-1} (\lambda - (l+1) - n) = 0$$

$$H'' = \sum_{n=0}^{\infty} a_n n(n-1) p^{n-2}$$

$$\sum_{n+1=1}^{\infty} a_{n+1} p^{n-1} \left\{ (n+1)(n+2(l+1)) + \sum_{n=0}^{\infty} a_n p^{n-1} (\lambda - (l+1) - n) \right\} = 0$$

$$(a_{n+1}) (n+1) (n+2(l+1)) = a_n \{ (l+1+n) - \lambda \}$$

$$\Rightarrow \frac{a_{n+1}}{a_n} = \frac{l+1+n-\lambda}{(n+1)(n+2(l+1))} \quad n r = \lambda - l - 1$$

$$\lambda = n r + l + 1$$

$$\Rightarrow \frac{a_{n+1}}{a_n} = \frac{n - (\lambda - l - 1)}{(n+1)(n+2(l+1))} \quad n: \text{bes Quantum zahl}$$

$$n = n r + l + 1$$

$$n \geq l + 1$$

$n=1$	$n_r=0$	$l=0$		$\} n^2$
$n=2$	$n_r=0$	$l=1$	$(2l+1)=3$	$\} n^2$
	$n_r=1$	$l=0$	$(2l+1)=1$	
$n=3$	$n_r=0$	$l=0$	$= \begin{matrix} 1 \\ 3 \\ 5 \end{matrix} \} 9$	$\} n^2$
	$n_r=1$	$l=1$		
	$n_r=2$	$l=2$		

Böhm 13

$$\nabla \cdot \vec{E} = \mu_0 \rho$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \frac{\mu_0}{c} \vec{J} + \frac{1}{c} \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \cdot \vec{E} = \mu_0 \rho \Rightarrow$$

$$\vec{B} = -\frac{1}{c} \int \vec{J}(\vec{x}') \frac{|\vec{x} - \vec{x}'|}{(\vec{x} - \vec{x}')^3} d\vec{x}'$$

$$\vec{B} = \frac{1}{c} \int \vec{J}(\vec{x}') \frac{1}{|\vec{x} - \vec{x}'|^2} d\vec{x}'$$

$$\vec{B} = \frac{1}{c} \nabla \times \int \vec{J} \frac{d\vec{x}}{|\vec{x} - \vec{x}'|}$$

$$\vec{B} = \nabla \times \underbrace{\frac{1}{c} \int \vec{J} \frac{d\vec{x}}{|\vec{x} - \vec{x}'|}}_{\vec{A}} \Rightarrow \vec{B} = \nabla \times \vec{A}$$

$$\vec{B} = \nabla \times \vec{A} \Rightarrow \vec{B}' = \nabla \times \vec{A}'$$

$$\Rightarrow \underline{\vec{A}' = \vec{A} - \nabla f} \Rightarrow \nabla \times (\vec{A} - \nabla f) = \nabla \times \vec{A} - \nabla \times (\nabla f) = \nabla \times \vec{A}$$

$$\text{i)} \nabla \cdot \vec{E} = \mu_0 \rho \Rightarrow \vec{E} = -\nabla \phi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t}$$

$$\Rightarrow \nabla \cdot (-\nabla \phi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t}) = \mu_0 \rho$$

$$\Rightarrow -\nabla^2 \phi - \frac{1}{c} \frac{\partial \nabla \cdot \vec{A}}{\partial t} = \mu_0 \rho \quad \nabla \cdot \vec{A} = 0 \Rightarrow \text{Coulomb gauge}$$

$$-\nabla^2 \phi = -\mu_0 \rho \Rightarrow \underline{\nabla^2 \phi = -\mu_0 \rho} \quad \text{Poisson's Eq}$$

$$\text{(ii)} \nabla \times \vec{B} = \frac{\mu_0}{c} \vec{J} + \frac{1}{c} \frac{\partial \vec{E}}{\partial t}$$

$$\Rightarrow \nabla \times \vec{B} = \frac{\mu_0}{c} \vec{J} + \frac{1}{c} \frac{\partial (-\nabla \phi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t})}{\partial t}$$

$$\nabla \times (\nabla \times \vec{A}) + \frac{1}{c} \frac{\partial}{\partial t} (\nabla \phi + \frac{1}{c} \frac{\partial \vec{A}}{\partial t})$$

$$\nabla^2 \vec{A} - \nabla (\nabla \cdot \vec{A}) + \frac{1}{c} \frac{\partial \nabla \phi}{\partial t} + \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = \frac{\mu_0}{c} \vec{J}$$

$$\nabla \left(\frac{1}{c} \frac{\partial \phi}{\partial t} - \nabla \cdot \vec{A} \right) = 0$$

$$\nabla \cdot \vec{A} = \frac{1}{c} \frac{\partial \phi}{\partial t}$$

$$\nabla \cdot \vec{E} = \mu_0 \rho \Rightarrow \nabla \cdot (-\nabla \phi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t}) = \mu_0 \rho$$

$$\Rightarrow (-\nabla^2 \phi - \frac{1}{c} \frac{\partial \nabla \cdot \vec{A}}{\partial t}) = \mu_0 \rho \Rightarrow -\nabla^2 \phi - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = \mu_0 \rho$$

$$\vec{E} = -\nabla \phi + \frac{1}{c} \frac{\partial \vec{A}}{\partial t} \Rightarrow \left(-\nabla^2 \phi + \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = \mu_0 \rho \right)$$

$\square^2$ De Lambert op

$$H \Rightarrow K.E + V \Rightarrow \frac{p^2}{2\mu} + \frac{e\phi}{0}$$

$$H\psi = E\psi \Rightarrow \frac{1}{2\mu} (\nabla^2 \psi) = E\psi$$

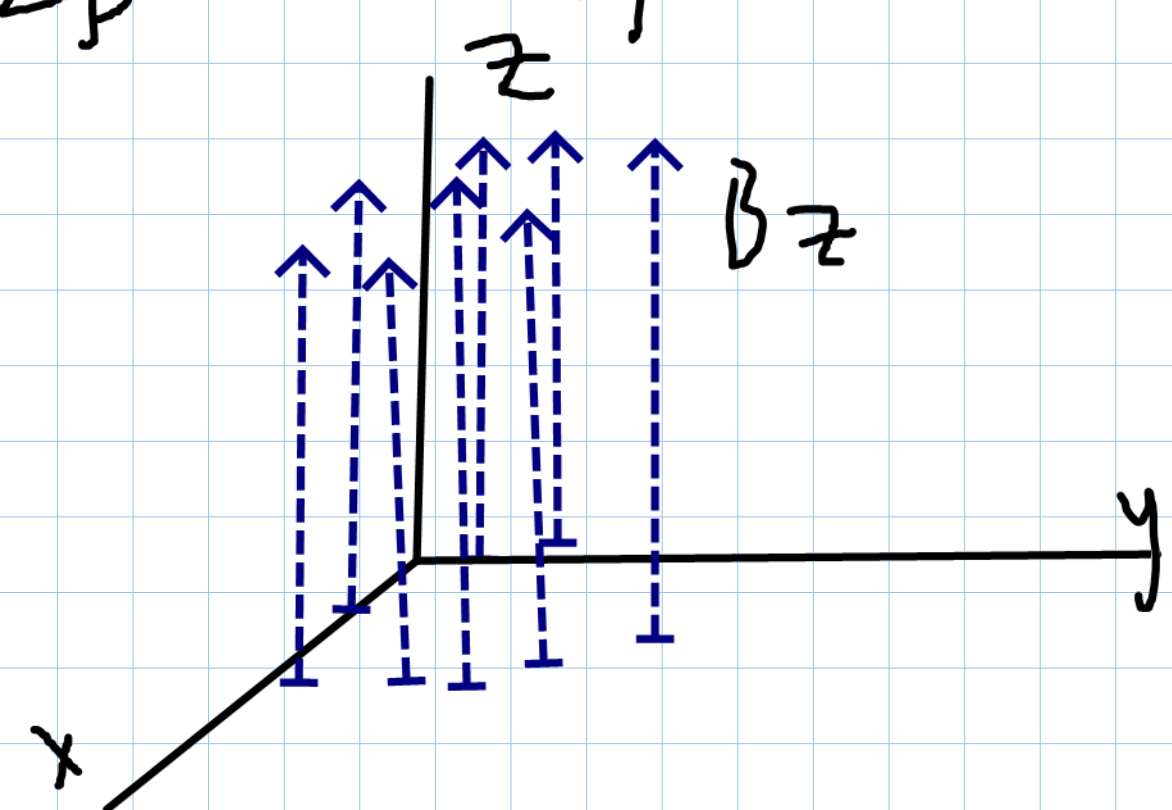
$$\frac{1}{2\mu} \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \psi = E\psi$$

$$\Rightarrow \frac{1}{2\mu} \left\{ \left(\frac{\hbar}{i} \nabla + \frac{e}{c} A \right) \left(\frac{\hbar}{i} \nabla \psi + \frac{e}{c} A \psi \right) \right\} = E\psi$$

$$\Rightarrow \frac{1}{2\mu} \left\{ -\hbar^2 \nabla^2 \psi + \frac{\hbar e}{i c} (\nabla A) \psi + \frac{\hbar e}{i c} A (\nabla \psi) + \frac{e^2}{c^2} A^2 \psi \right\}$$

$$\Rightarrow \frac{1}{2\mu} \left\{ -\hbar^2 \nabla^2 \psi + 2 \frac{\hbar e}{i c} A (\nabla \psi) + \frac{e^2}{c^2} A^2 \psi \right\}$$

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi + \frac{\hbar e}{i \mu c} A (\nabla \psi) + \frac{e^2}{2\mu c^2} A^2 \psi = E\psi$$



$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi + \frac{\hbar e}{i \mu c} A (\nabla \psi) + \frac{e^2}{2\mu c^2} A^2 \psi = E\psi$$

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi + \frac{\hbar e}{i \mu c} \left(-\frac{1}{2} \right) (r \times B) (\nabla \psi) + \frac{e^2}{2\mu c^2} \left(\frac{B^2}{4} \right) (x^2 + y^2)$$

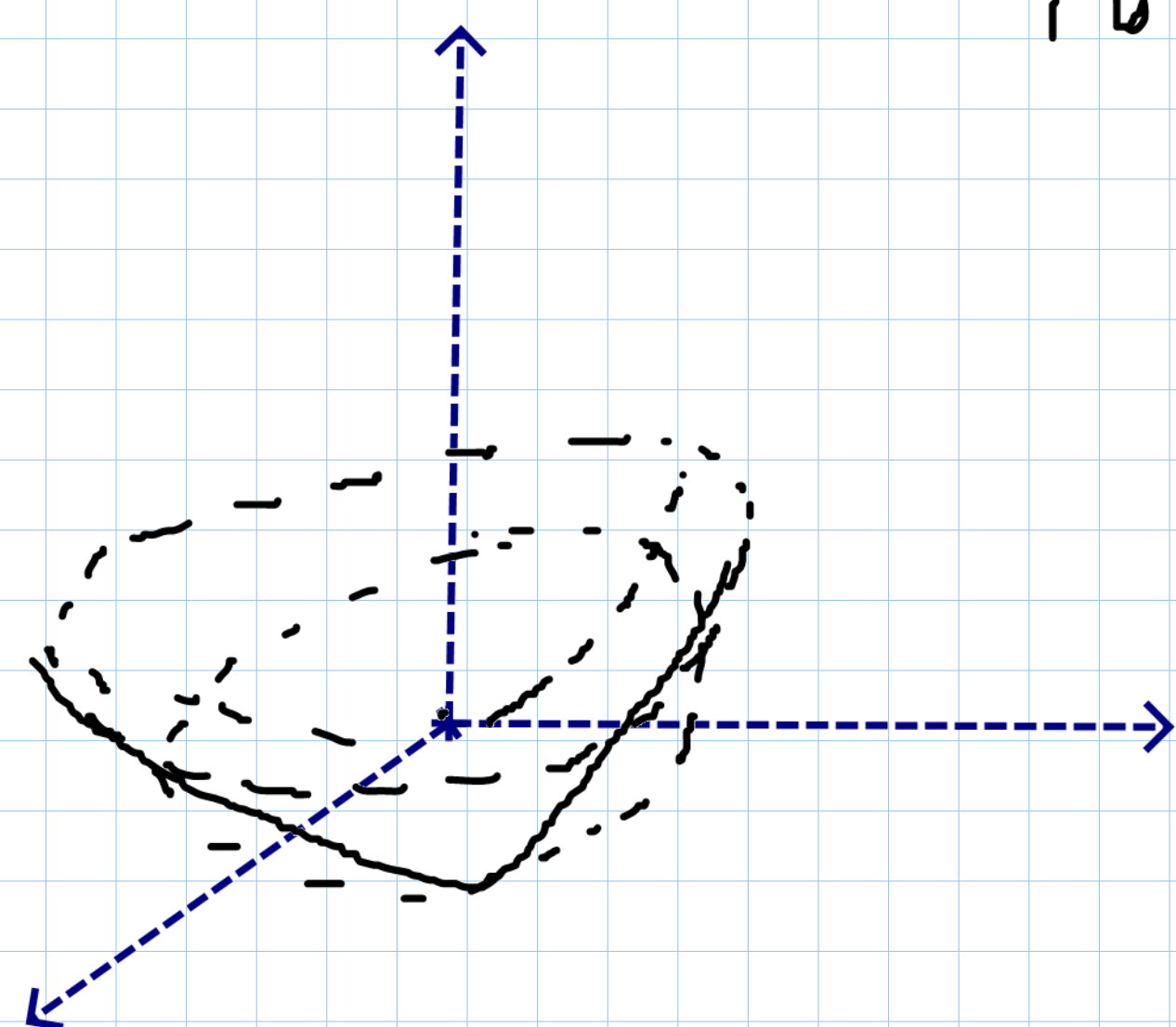
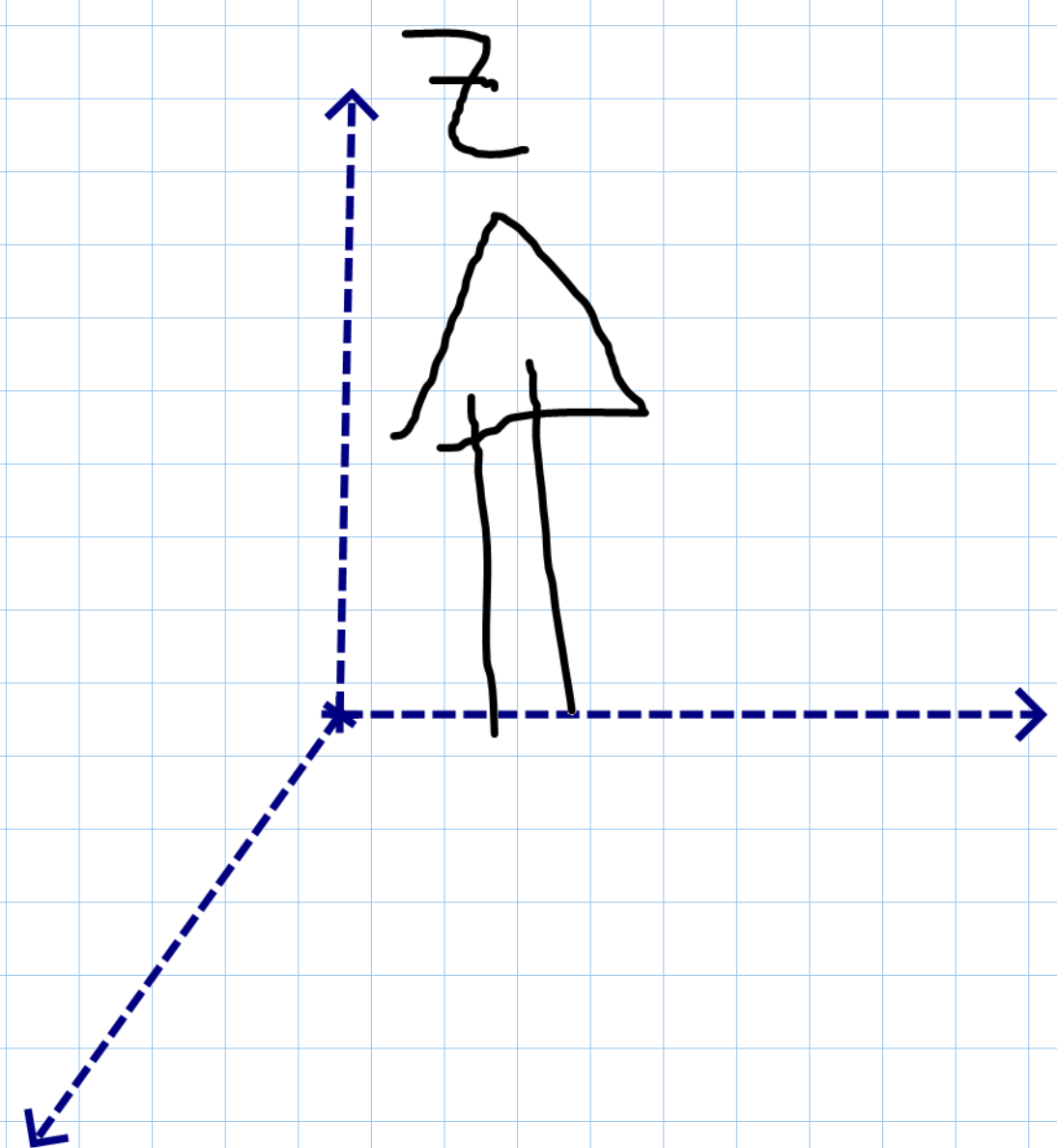
$$\Rightarrow -\frac{\hbar^2}{2\mu} \nabla^2 \psi + \frac{\hbar e}{2i \mu c} (r \times \nabla) B \psi + \frac{e^2 B^2}{8\mu c^2} (x^2 + y^2)$$

$$\Rightarrow -\frac{\hbar^2}{2\mu} \nabla^2 \psi + \frac{e B}{2\mu c} (r \times \frac{\hbar}{i} \nabla) \psi + \frac{e^2 B^2}{8\mu c^2} (x^2 + y^2) = 0$$

L_z

$$\Rightarrow -\frac{\hbar^2}{2\mu} \nabla^2 \psi + \frac{e B}{2\mu c} L_z \psi + \frac{e^2 B^2}{8\mu c^2} (x^2 + y^2) = 0$$

$$L_z = \frac{\hbar}{i} \frac{\partial}{\partial \phi}$$



$$A = -\frac{1}{2} (r \times B) \Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ B_x & B_y & B_z \end{vmatrix}$$

$$(r \times B) = \hat{i}(yB_z - zB_y) - \hat{j}(xB_z - zB_x) + \hat{k}(xB_y - yB_x)$$

$$yB_z \hat{i} - xB_z \hat{j} \Rightarrow B_z = |B| = B$$

$$\star \Rightarrow B_z(-y, +x, 0) = \text{coulomb avg}$$

$$\text{Larmor avg} \Rightarrow \frac{B(-y, 0, 0)}{2} \text{ very } \frac{B(0, +x, 0)}{2}$$

$$\Rightarrow A = -\frac{1}{2} (r \times B) = \boxed{-\frac{B}{2} (y, -x, 0) = A}$$

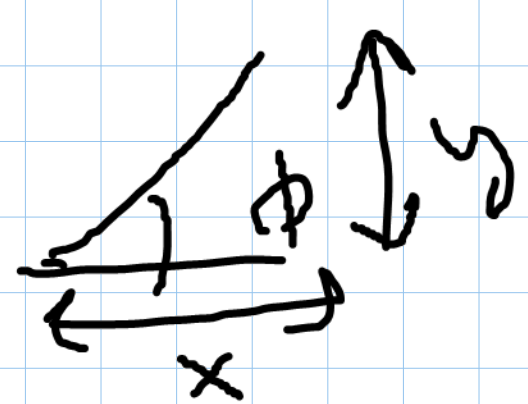
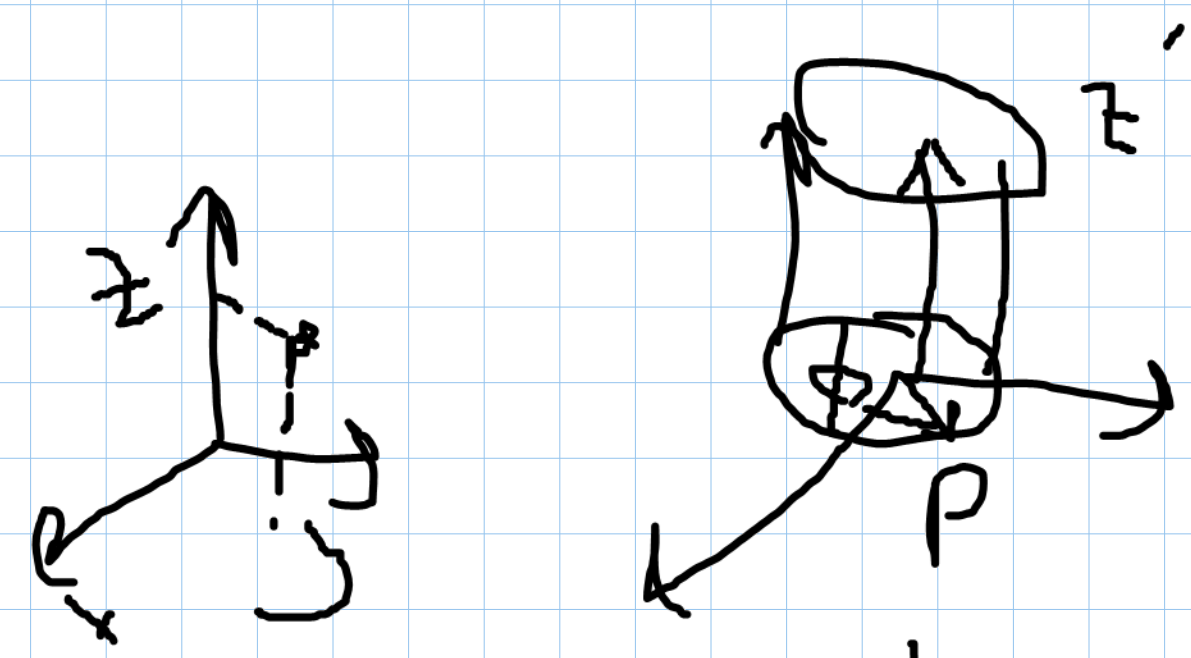
$$A^2 = \frac{B^2}{4} (x^2 + y^2)$$

$$\frac{e B}{2\mu c} = \omega_L \text{ Larmor Frekuensi}$$

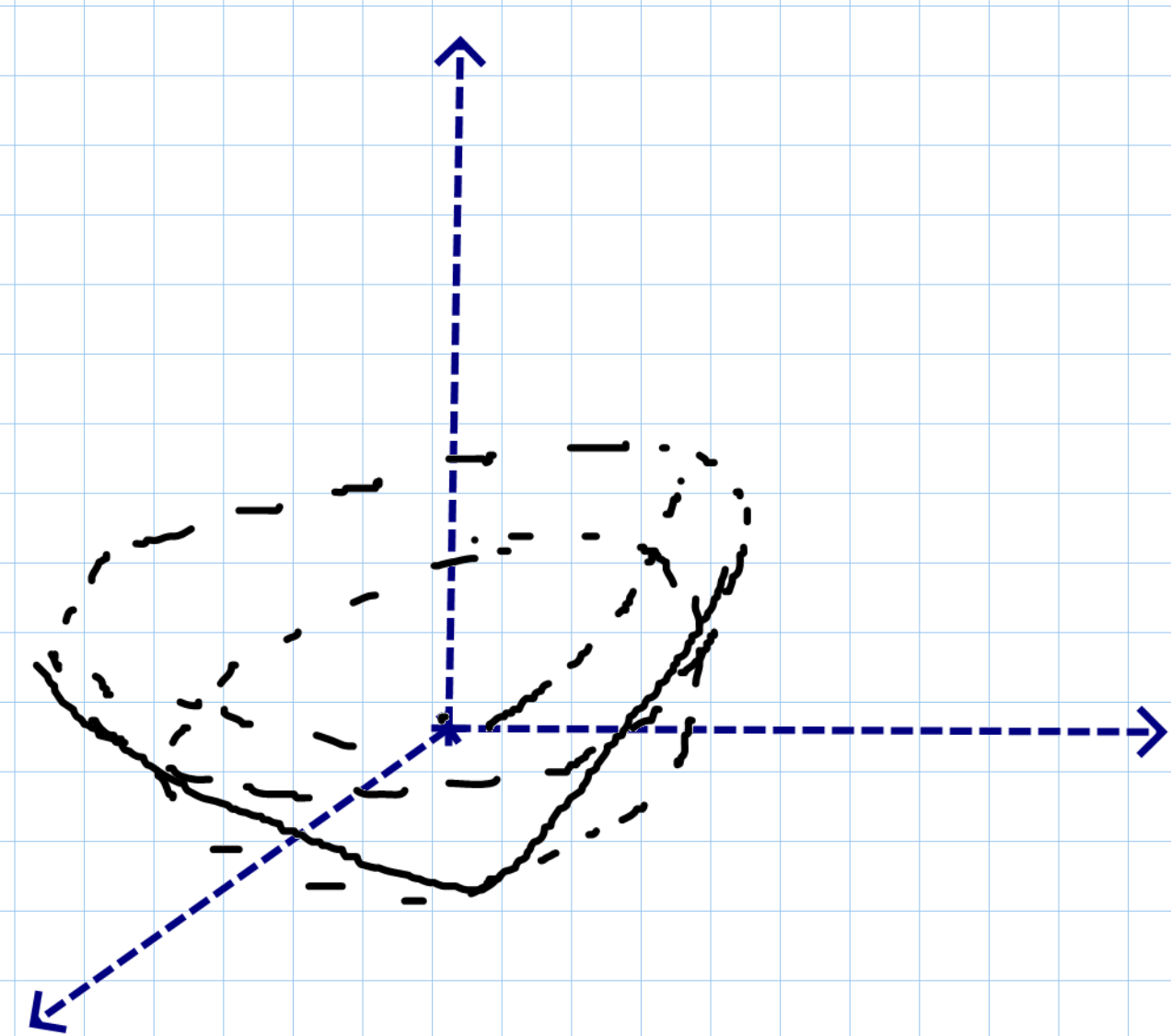
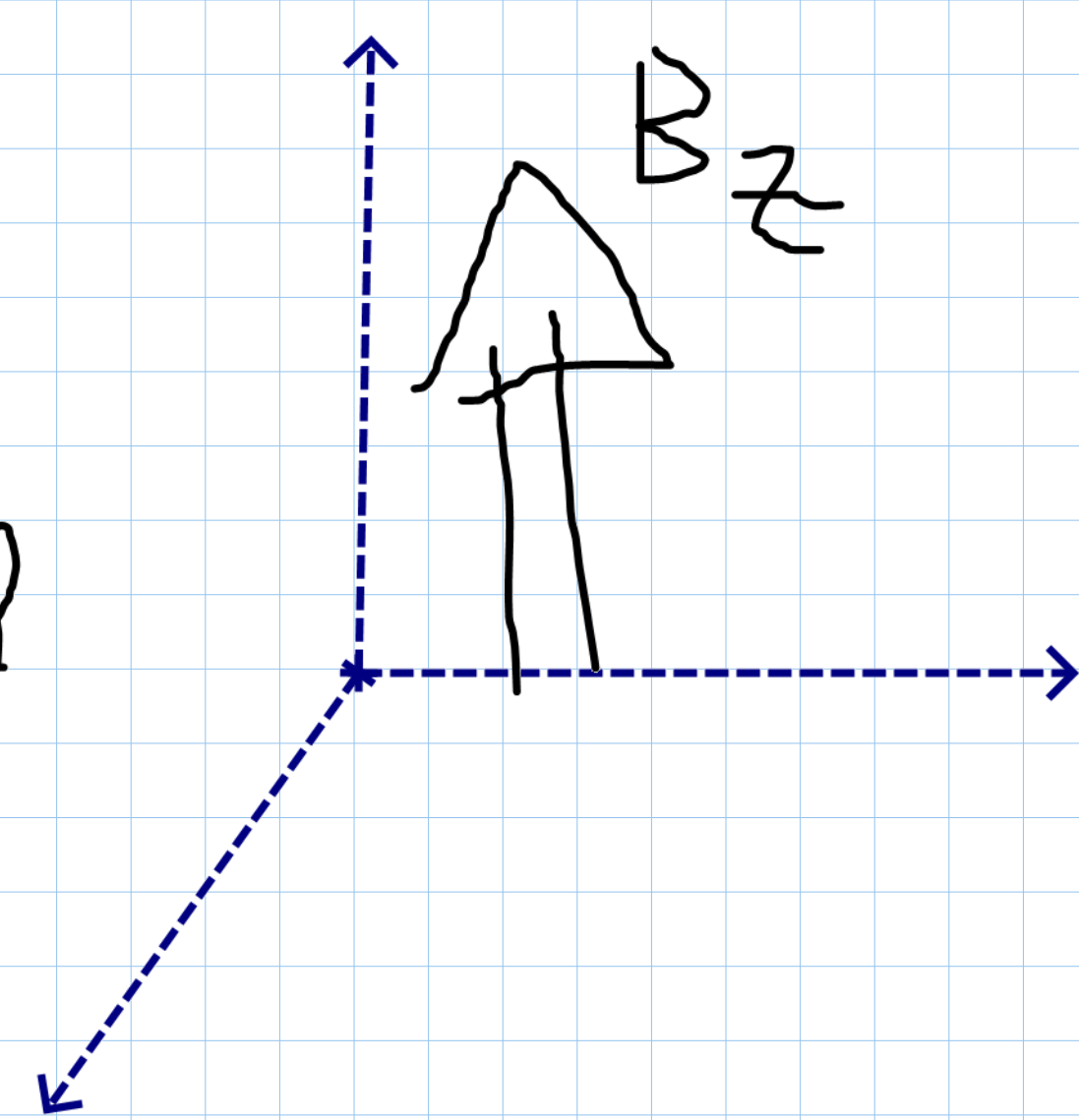
$$\frac{-\hbar^2}{2\mu} \nabla^2 \psi + \frac{eB\hbar}{2\mu c} L_z + \frac{e^2 B^2}{8\mu c^2} (x^2 + y^2) = 0 \quad \star$$

$$L_z = \frac{\hbar}{i} \frac{\partial}{\partial \varphi}$$

Kartezyden + silindric



$$\tan \theta = \frac{y}{x} \Rightarrow \theta = \arctan \frac{y}{x}$$



$$\rho = (x^2 + y^2)^{1/2} \quad z = z \quad \varphi = \arctan(y/x)$$

$$\frac{\partial}{\partial x} = \frac{\partial \rho}{\partial x} \frac{\partial}{\partial \rho} + \frac{\partial \varphi}{\partial x} \frac{\partial}{\partial \varphi} + \frac{\partial z}{\partial x} \frac{\partial}{\partial z}$$

$$\frac{\partial}{\partial y} = \frac{\partial \rho}{\partial y} \frac{\partial}{\partial \rho} + \frac{\partial \varphi}{\partial y} \frac{\partial}{\partial \varphi} + \frac{\partial z}{\partial y} \frac{\partial}{\partial z}$$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial z}$$

$$\frac{\partial}{\partial x} = \frac{\partial \rho}{\partial x} \frac{\partial}{\partial \rho} + \frac{\partial \varphi}{\partial x} \frac{\partial}{\partial \varphi} \Rightarrow \frac{\partial}{\partial x} = \frac{d(x^2 + y^2)^{1/2}}{dx} \frac{\partial}{\partial \rho} + \frac{d(\arctan(y/x))}{dx} \frac{\partial}{\partial \varphi}$$

$$\frac{\partial}{\partial y} = \frac{\partial \rho}{\partial y} \frac{\partial}{\partial \rho} + \frac{\partial \varphi}{\partial y} \frac{\partial}{\partial \varphi} \Rightarrow \frac{\partial}{\partial y} = \frac{d(x^2 + y^2)^{1/2}}{dy} \frac{\partial}{\partial \rho} + \frac{d(\arctan(y/x))}{dy} \frac{\partial}{\partial \varphi}$$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial z}$$

$$\Rightarrow \frac{\partial}{\partial x} = \frac{1}{2} (x^2 + y^2)^{-1/2} \cdot 2x \frac{\partial}{\partial \rho} - \frac{1}{x^2} \frac{y}{1 + \frac{y^2}{x^2}} \frac{\partial}{\partial \varphi}$$

$$\Rightarrow \frac{\partial}{\partial x} = \frac{x}{(x^2 + y^2)^{1/2}} \frac{\partial}{\partial \rho} - \frac{y}{x^2 + y^2} \frac{\partial}{\partial \varphi}$$

$$\Rightarrow \frac{\partial}{\partial x} = \cos \varphi \frac{\partial}{\partial \rho} - \frac{\sin \varphi}{\rho} \frac{\partial}{\partial \varphi}$$

$$\frac{\partial}{\partial y} = \sin \varphi \frac{\partial}{\partial \rho} + \frac{\cos \varphi}{\rho} \frac{\partial}{\partial \varphi}$$

$$\Rightarrow \frac{\partial^2}{\partial x^2} = \left(\cos \varphi \frac{\partial}{\partial \rho} - \frac{\sin \varphi}{\rho} \frac{\partial}{\partial \varphi} \right) \left(\cos \varphi \frac{\partial}{\partial \rho} - \frac{\sin \varphi}{\rho} \frac{\partial}{\partial \varphi} \right) =$$

$$\Rightarrow \cos^2 \varphi \frac{\partial^2}{\partial \rho^2} + \cos \varphi \frac{\sin \varphi}{\rho^2} \frac{\partial}{\partial \varphi} + \frac{\sin^2 \varphi}{\rho} \frac{\partial}{\partial \rho} + \frac{\sin \varphi \cos \varphi}{\rho^2} \frac{\partial}{\partial \varphi} + \frac{\sin^2 \varphi}{\rho^2} \frac{\partial^2}{\partial \varphi^2}$$

$$\frac{\partial^2}{\partial y^2} \Rightarrow \left(\sin \varphi \frac{\partial}{\partial \rho} + \frac{\cos \varphi}{\rho} \frac{\partial}{\partial \varphi} \right) \left(\sin \varphi \frac{\partial}{\partial \rho} + \frac{\cos \varphi}{\rho} \frac{\partial}{\partial \varphi} \right) =$$

$$\Rightarrow \sin^2 \varphi \frac{\partial^2}{\partial \rho^2} - \frac{\sin \varphi \cos \varphi}{\rho^2} \frac{\partial}{\partial \varphi} + \frac{\cos^2 \varphi}{\rho} \frac{\partial}{\partial \rho} - \frac{\cos \varphi \sin \varphi}{\rho^2} \frac{\partial}{\partial \varphi} + \frac{\cos^2 \varphi}{\rho^2} \frac{\partial^2}{\partial \varphi^2}$$

$$\frac{\partial}{\partial z^2} = \frac{\partial}{\partial z^2}$$

$$(\arctan u)' = \frac{u'}{1+u^2}$$

$$\cos^2 \varphi \frac{d^2}{d\rho^2} + \frac{\cos \varphi \sin \varphi}{\rho^2} \frac{d}{d\varphi} + \frac{\sin^2 \varphi}{\rho} \frac{d}{d\rho} + \frac{\sin \varphi \cos \varphi}{\rho^2} \frac{d}{d\varphi} + \frac{\sin^2 \varphi}{\rho^2} \frac{d^2}{d\varphi^2}$$

$$\sin^2 \varphi \frac{d^2}{d\rho^2} - \frac{\sin \varphi \cos \varphi}{\rho^2} \frac{d}{d\varphi} + \frac{\cos^2 \varphi}{\rho} \frac{d}{d\rho} - \frac{\cos \varphi \sin \varphi}{\rho^2} \frac{d}{d\varphi} + \frac{\cos^2 \varphi}{\rho^2} \frac{d^2}{d\varphi^2}$$

$$\frac{d^2}{d\rho^2} + \frac{1}{\rho} \frac{d}{d\rho} + \frac{1}{\rho^2} \frac{d^2}{d\varphi^2} + \frac{d^2}{dz^2} = \nabla^2_{cyl}$$

$$\frac{-\hbar^2}{2\mu} \nabla^2 \psi + \frac{eB\hbar}{2\mu c} L_z + \frac{e^2 B^2}{8\mu c^2} (x^2 + y^2) = 0 \quad \star$$

$$-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{d\rho^2} + \frac{1}{\rho} \frac{d}{d\rho} + \frac{1}{\rho^2} \frac{d^2}{d\varphi^2} + \frac{d^2}{dz^2} \right) \psi + \frac{eB\hbar}{2\mu c} \frac{d}{d\varphi} \psi + \frac{e^2 B^2}{8\mu c^2} \rho^2 = 0$$

↓ Buradan azimutal bir alt + ek: azimutal dnyl sonucu verir.

$$\frac{\partial^{n+1}}{\partial n} = \frac{n - (n-1-1)}{(n+1)(n+2(1+1))} \quad n: \text{baş kuantum sayısı}$$

$$n = n_r + l + 1$$

$$n \geq 0 + 1$$

Soru $\frac{1}{2\mu} \left(p + \frac{e}{c} A \right) \left(p + \frac{e}{c} A \right) \psi = E \psi \quad \vec{A} = \vec{B}(-y, 0, 0) \Rightarrow \vec{A} = B(-y\hat{x} + 0\hat{y} + 0\hat{z})$

$$\vec{A} = -B(y, 0, 0)$$

$$A_x = -By$$

$$\Rightarrow \frac{1}{2\mu} \left((p_x + \frac{e}{c} A_x) (p_x + \frac{e}{c} A_x) + p_y^2 + p_z^2 \right) \psi = E \psi$$

$$\Rightarrow \frac{1}{2\mu} \left\{ \left(p_x - \frac{e}{c} By \right) \left(p_x - \frac{e}{c} By \right) + p_y^2 + p_z^2 \right\} \psi = E \psi \quad e^{ik_x x} \cdot e^{ik_z z} \cdot y_{ly} = \psi$$

$$\Rightarrow \frac{1}{2\mu} \left(\underbrace{\left(\frac{\hbar}{i} \frac{d}{dx} - \frac{e}{c} By \right) \left(\frac{\hbar}{i} \frac{d}{dx} - \frac{e}{c} By \right)}_{(i)} e^{ik_x x + ik_z z} y_{ly} + \underbrace{\left(\frac{\hbar}{i} \frac{d}{dy} \right) \left(\frac{\hbar}{i} \frac{d}{dy} \right)}_{(ii)} e^{ik_x x + ik_z z} y_{ly} + \underbrace{\left(\frac{\hbar}{i} \frac{d}{dz} \right) \left(\frac{\hbar}{i} \frac{d}{dz} \right)}_{(iii)} e^{ik_x x + ik_z z} y_{ly} \right)$$

$$\Rightarrow \frac{1}{2\mu} \left(\left(\frac{\hbar}{i} \frac{d}{dx} - \frac{e}{c} By \right) \cdot \left(\hbar k_x e^{ik_x x + ik_z z} y_{ly} - \frac{e}{c} By e^{ik_x x + ik_z z} y_{ly} \right) \right)$$

$$(i) \Rightarrow \frac{1}{2\mu} \left\{ \left(\hbar^2 k_x^2 e^{ik_x x + ik_z z} y_{ly} - \frac{e\hbar}{c} By k_x e^{ik_x x + ik_z z} y_{ly} - \frac{e}{c} By \hbar k_x e^{ik_x x + ik_z z} y_{ly} + \frac{e^2}{c^2} B^2 y^2 e^{ik_x x + ik_z z} y_{ly} \right) \right\}$$

$$\frac{1}{2\mu} \left\{ \left(\hbar^2 k_x^2 - \frac{2e\hbar}{c} By k_x + \frac{e^2}{c^2} B^2 y^2 \right) e^{ik_x x + ik_z z} y_{ly} \right\}$$

$$(ii) \Rightarrow \frac{\hbar^2}{2\mu} \frac{d^2}{dy^2} y_{ly} e^{ik_x x + ik_z z} \quad (iii) \Rightarrow \frac{\hbar^2 k_z^2}{2\mu} e^{ik_x x + ik_z z} y_{ly}$$

$$\frac{1}{2\mu} \left\{ \left(\hbar^2 k_x^2 - \frac{2e\hbar}{c} By k_x + \frac{e^2}{c^2} B^2 y^2 \right) e^{ik_x x + ik_z z} y_{ly} - \frac{\hbar^2}{2\mu} \frac{d^2}{dy^2} y_{ly} e^{ik_x x + ik_z z} + \frac{\hbar^2 k_z^2}{2\mu} y_{ly} e^{ik_x x + ik_z z} \right\} = E y_{ly} e^{ik_x x + ik_z z}$$

$$\frac{1}{2\mu} \left(\hbar^2 k_x^2 - \frac{2eBy\hbar k_x}{c} + \frac{e^2 B^2 y^2}{c^2} \right) e^{ik_x x + ik_z z} \psi(y) - \frac{\hbar^2}{2\mu} \frac{d^2 \psi}{dy^2} e^{ik_x x + ik_z z} + \frac{\hbar^2 k_z^2}{2\mu} e^{ik_x x + ik_z z} \psi(y) = e^{ik_x x + ik_z z} \psi(y) E$$

$$e^{ik_x x + ik_z z}$$

$$\Rightarrow \frac{1}{2\mu} \left(\hbar^2 k_x^2 - \frac{2eBy\hbar k_x}{c} + \frac{e^2 B^2 y^2}{c^2} \right) - \frac{\hbar^2}{2\mu} \frac{d^2 \psi}{dy^2} = (E - \frac{\hbar^2 k_z^2}{2\mu}) \psi$$

$$\frac{\hbar^2 k_x^2}{2\mu} - \frac{eBy\hbar k_x}{\mu c} + \frac{e^2 B^2 y^2}{2\mu c^2} \Rightarrow \frac{1}{2\mu} \left\{ \hbar^2 k_x^2 - \frac{2eBy\hbar k_x}{c} + \frac{e^2 B^2 y^2}{c^2} \right\}$$

$$\frac{1}{2\mu} \left(\hbar k_x - \frac{eBy}{c} \right)^2$$

$$\frac{1}{2\mu} \left(\hbar k_x - \frac{eBy}{c} \right)^2 - \frac{\hbar^2}{2\mu} \frac{d^2 \psi}{dy^2} = \left(E - \frac{\hbar^2 k_z^2}{2\mu} \right) \psi$$

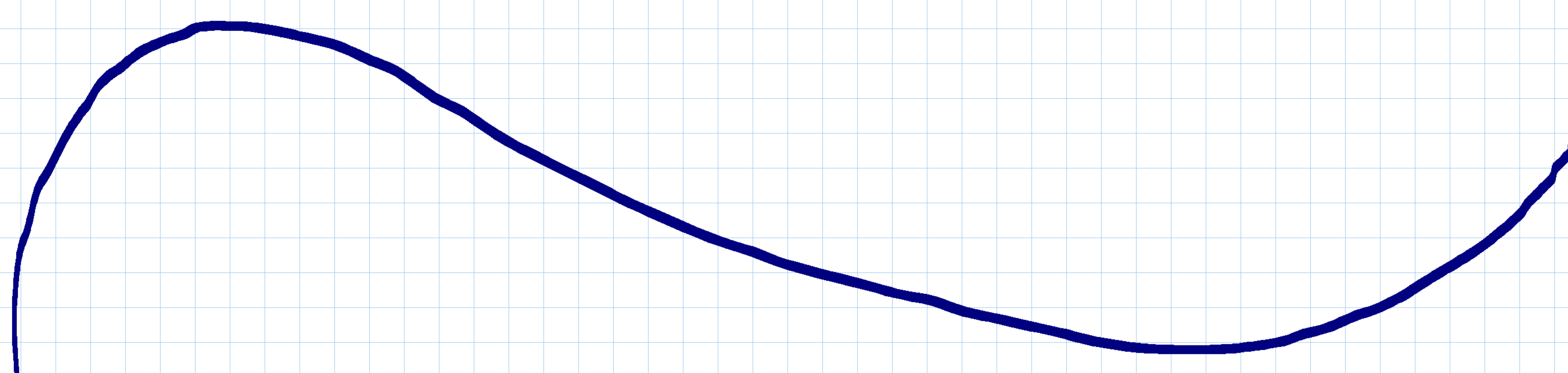
$$\omega_L = \frac{eB}{\mu c}$$

$$\omega_L^2 = \frac{e^2 B^2}{\mu^2 c^2}$$

$$\mu \frac{\hbar^2}{\mu^2} \left(\hbar k_x - \frac{eBy}{c} \right)^2 \psi - \frac{\hbar^2}{2\mu} \frac{d^2 \psi}{dy^2} = \left(E - \frac{\hbar^2 k_z^2}{2\mu} \right) \psi$$

$$\frac{\mu}{2} \left(\hbar k_x - \frac{eBy}{\mu c} \right)^2 \psi \Rightarrow \frac{\mu}{2} \left(\frac{eB}{\mu c} \right)^2 \left(\hbar k_x \left(\frac{\mu c}{eB} \right) - y \right)^2 \psi$$

$$\Rightarrow -\frac{\mu}{2} \omega_L^2 (y - y_0)^2 \psi - \frac{\hbar^2}{2\mu} \frac{d^2 \psi}{dy^2} = E' \psi$$



5. Bir protonun Coulomb alanı içindeki bir elektron,

$$\frac{1}{6} [4\psi_{100}(\vec{r}) + 3\psi_{211}(\vec{r}) - \psi_{210}(\vec{r}) + \sqrt{10}\psi_{21-1}(\vec{r})]$$

dalga fonksiyonu ile betimlenen bir durumdur. Elektronun, hidrojen atomunun taban

(a) Enerjinin beklenen değeri nedir?

(b) $\vec{L}^2$ 'nin beklenen değeri nedir?(c) L_z 'nin beklenen değeri nedir?

6. Bir protonun Coulomb alanı içindeki bir elektron,

$$\psi(\vec{r}) = \left(\frac{\alpha}{\sqrt{\pi}} \right)^{3/2} e^{-\alpha^2 r^2/2}$$

dalga fonksiyonu ile betimlenen bir durumdur. Elektronun, hidrojen atomunun taban

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7. Bir elektron, hidrojen atomunun $n=2$, $\ell=1$, $m=0$ durumundadır. Elektrona momentum uzayındaki dalga fonksiyonu nedir?8. $f(\vec{r}, \vec{p})$ 'nin herhangi bir duraklı durumdaki beklenen değeri bir sabittir.

Bir

$$H = \vec{p}^2/2m + V(r)$$

Soru 5

d) $\Rightarrow \langle E \rangle = ? \Rightarrow \langle E \rangle = \langle \psi | H | \psi \rangle = \langle E \rangle \quad H \psi_n = E_n \psi_n$

$\Rightarrow \frac{1}{6} [4\psi_{100} + 3\psi_{211} - \psi_{210} + \sqrt{10}\psi_{21-1}]$

$\Rightarrow \langle \psi_n | H | \psi_n \rangle = \langle \psi_n | E_n | \psi_n \rangle = E_n \langle \psi_n | \psi_n \rangle = E_n \delta_{nn}$

$\Rightarrow \frac{1}{6} \cdot \frac{1}{6} [16 \langle \psi_{100} | H | \psi_{100} \rangle + 9 \langle \psi_{211} | H | \psi_{211} \rangle + \langle \psi_{210} | H | \psi_{210} \rangle + 10 \langle \psi_{21-1} | H | \psi_{21-1} \rangle]$

$\Rightarrow \frac{1}{36} [16 E_1 \langle \psi_{100} | \psi_{100} \rangle + 9 E_2 \langle \psi_{211} | \psi_{211} \rangle + E_2 \langle \psi_{210} | \psi_{210} \rangle + 10 E_2 \langle \psi_{21-1} | \psi_{21-1} \rangle]$

$\Rightarrow \frac{16}{9} E_1 + \frac{1}{4} E_2 + \frac{1}{36} E_2 + \frac{10}{36} E_2 \Rightarrow \frac{16}{36} E_1 + \frac{20}{36} E_2 //$

(b, c) Benzer şekilde yapılır

Soru 6

$\left(\frac{\alpha}{\sqrt{\pi}} \right)^{3/2} e^{-\alpha^2 r^2/2} \quad T.D? = \langle r \rangle = ?$

$$\langle \psi | r | \psi \rangle = \langle r \rangle$$

$$\Rightarrow \int d^3r \psi^* r \psi = \int d^3r \left(\frac{\alpha}{\sqrt{\pi}} \right)^3 e^{-\alpha^2 r^2} \cdot r =$$

$$\Rightarrow \left(\frac{\alpha}{\sqrt{\pi}} \right)^3 4\pi \int_0^\infty r^2 dr \cdot r e^{-\alpha^2 r^2}$$

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$$

$$\alpha^2 r^2 = t \Rightarrow 2r \alpha^2 dr = dt \Rightarrow r dr = \frac{dt}{2\alpha^2}$$

$$r^2 = \frac{t}{\alpha^2}$$

$$\Rightarrow \left(\frac{\alpha^3}{\pi\sqrt{\pi}} \right) 4\pi \cdot \frac{1}{2\alpha^2} \int_0^\infty dt \cdot \frac{t}{\alpha^2} e^{-t}$$

$$\Rightarrow \frac{2}{\alpha\sqrt{\pi}} \int_0^\infty dt t e^{-t} \quad \Gamma(2) = 1 \Rightarrow \alpha = \frac{2}{a_0\sqrt{\pi}}$$